

Jeho Lee

Ph.D. Candidate (expected graduation: Feb 2027)
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RESEARCH INTERESTS

- On-Device AI, Systems for ML, Mobile Systems

EDUCATION

Yonsei University - Seoul, Republic of Korea Ph.D. Student in Computer Science and Engineering (Advisor: Hojung Cha)	Mar 2020 – Present (Exp. Feb 2027)
Ajou University - Suwon, Republic of Korea B.S. in Computer Engineering	Mar 2015 – Feb 2020 GPA: 4.0/4.5

PUBLICATIONS

15. viNPU: Optimizing Vision Transformer Inference on Mobile NPUs
Jeho Lee, Gunjoong Kim, Chanyoung Jung, Jaehee Kim, Seonghoon Park, Hojung Cha
European Conference on Computer Systems (**EuroSys 2026**)
14. 3DGS on Your Phone: Towards Fully Immersive Mobile Volumetric Video Streaming
Gunjoong Kim*, Seonghoon Park*, **Jeho Lee**, Chanyoung Jung, Hyungchol Jun, Hojung Cha
ACM GetMobile: Mobile Computing and Communications (**GetMobile**)
13. VFM in Your Hands: Optimizing Real-Time Scene Understanding for Mobile Augmented Reality
Jeho Lee*, Chanyoung Jung*, Gunjoong Kim, Jiwon Kim, Seonghoon Park, Hojung Cha
ACM GetMobile: Mobile Computing and Communications (**GetMobile**)
12. Phoenix: Thermal-Aware On-device Inference of Multi-Instance DNNs for Mobile Video Applications
Seunghyeok Jeon, Jiwon Kim, **Jeho Lee**, Hojung Cha
ACM Transactions on Embedded Computing Systems (**TECS**)
11. Vega: Fully Immersive Mobile Volumetric Video Streaming with 3D Gaussian Splatting
Gunjoong Kim*, Seonghoon Park*, **Jeho Lee**, Chanyoung Jung, Hyungchol Jun, Hojung Cha
ACM International Conference on Mobile Computing and Networking (**MobiCom 2025**)
10. EOS: Energy-Optimized Super-Resolution on Mobile Devices for Live 360-Degree Videos
Seonghoon Park, Minchan Kim, Hyejin Park, **Jeho Lee**, Jiwon Kim, Hojung Cha
ACM International Conference on Mobile Computing and Networking (**MobiCom 2025**)
9. Ember: Task Wakeup Sequence-Based Energy Optimization for Mobile Web Browsing
Seonghoon Park, Jiwon Kim, **Jeho Lee**, Hojung Cha
ACM SIGBED International Conference on Embedded Software (**EMSOFT 2025**)
8. SecureRide: Detecting Safety-threatening Behavior of E-Scooters Using Battery Information
Jiwon Kim, Geon Kim, **Jeho Lee**, Thiemo Voigt, Hojung Cha
ACM SIGBED International Conference on Embedded Software (**EMSOFT 2025**)
7. ARIA: Optimizing Vision Foundation Model Inference on Heterogeneous Mobile Processors for Augmented Reality
Jeho Lee*, Chanyoung Jung*, Gunjoong Kim, Jiwon Kim, Seonghoon Park, Hojung Cha
ACM International Conference on Mobile Systems, Applications and Services (**MobiSys 2025**)
Best Paper Award (2 out of 233 submissions)

6. Towards Accurate, Adaptive, and Real-time Machine Perception on Resource-constrained Platforms
Jeho Lee
ACM International Conference on Mobile Systems, Applications and Services (**MobiSys 2025 Rising Star**)

5. Poster: Mixture of Class-aware Experts for Efficient AIoT Inference
Hyemin Jeong, **Jeho Lee**, Seunghyeok Jeon, Hojung Cha
ACM International Conference on Mobile Systems, Applications and Services (**MobiSys 2025 Poster**)

4. Panopticus: Omnidirectional 3D Object Detection on Resource-constrained Edge Devices
Jeho Lee, Chanyoung Jung, Jiwon Kim, Hojung Cha
ACM International Conference on Mobile Computing and Networking (**MobiCom 2024**)

3. Vulture: Cross-Device Web Experience with Fine-Grained Graphical User Interface Distribution
Seonghoon Park, **Jeho Lee**, Yonghun Choi, Hojung Cha
IEEE Conference on Computer Communications (**INFOCOM 2024**)

2. OmniLive: Super-Resolution Enhanced 360° Video Live Streaming for Mobile Devices
Seonghoon Park*, Yeonwoo Cho*, Hyungchol Jun, **Jeho Lee**, Hojung Cha
ACM International Conference on Mobile Systems, Applications and Services (**MobiSys 2023**)

1. Crow API: Cross-device I/O Sharing in Web Applications
Seonghoon Park, **Jeho Lee**, Hojung Cha
IEEE Conference on Computer Communications (**INFOCOM 2023**)

PROJECTS

Development of On-device DNN Inference System for Real-time 3D Perception with Mobile 360-degree Camera	
National Research Foundation of Korea (NRF)	May 2024 – Present
Task Relation Graph Prediction based on RNN	
Samsung Electronics, Republic of Korea	Mar 2023 – Feb 2024

AWARDS AND HONORS

Best Paper Award , ACM MobiSys 2025 (2 out of 233 submissions)	Jun 2025
Rising Star , ACM MobiSys 2025	Jun 2025
SIGMOBILE Student Travel Grant , ACM MobiSys 2025	May 2025

PATENTS

3. I/O Sharing Device and Method
Seonghoon Park, **Jeho Lee**, and Hojung Cha
Patent No. 10-2823808 (Republic of Korea; granted Jun. 18, 2025)

2. Method for Omnidirectional 3D Object Detection, Program Performing the Method, and Computing Device Executing the Program
Jeho Lee, Chanyoung Jung, Seonghoon Park, Hyungchol Jun, and Hojung Cha
Patent Pending, Patent Application No. 10-2024-0120347 (Republic of Korea; filed Sep. 04, 2024)

1. System and Operating Method for Cross-Device Experiences using In-Browser Virtual Proxy
Seonghoon Park, **Jeho Lee**, and Hojung Cha
Patent Pending, Patent Application No. 10-2024-0112156 (Republic of Korea; filed Aug. 21, 2024)

SKILLS

Languages: English (Fluent), Korean (Native)
Programming: Python, C/C++, Java, JavaScript
Frameworks: Android, Qualcomm Neural Processing SDK (QNN), TensorRT, PyTorch, TensorFlow Lite, MMDetection3D, ROS, Node.js
Hardware Platforms: Qualcomm Snapdragon SoCs (with Hexagon NPUs), NVIDIA Jetson SoCs

TEACHING EXPERIENCE

Guest Lecturer, Yonsei University On-Device Intelligence for 3D Perception: Challenges and Innovations	Spring 2024
Teaching Assistant, Yonsei University Operating Systems (CSI3101)	Spring 2023
Teaching Assistant, Yonsei University Introduction to Computer Science (CSI2106)	Fall 2022

MENTORING EXPERIENCE

Dasol Yoon , Master student at Yonsei Univ., working on an on-device AI project	2025 – Present
Jaehye Kim , Master student at Yonsei Univ., a EuroSys 2026 paper	2025 – Present
Hyemin Jeong , Master student at Yonsei Univ., a MobiSys 2025 poster	2025
Gunjoong Kim , Master student at Yonsei Univ., MobiSys 2025/MobiCom 2025 papers	2024 – 2025
Chanyoung Jung , Master student at Yonsei Univ., MobiSys 2025/MobiCom 2025 papers	2024 – 2025
Chanyoung Jung , Undergraduate student at Yonsei Univ., a MobiCom 2024 paper	2022 – 2023
Software Capstone Design , Yonsei Univ.	Spring 2021, Fall 2021, Spring 2022, Spring 2023, Fall 2023

ACADEMIC SERVICES

Student Volunteer , MobiSys 2024
